

TRIBE v2 Evaluation

Sprint 1 — IBC Data Acquisition & Parcellation

Project	TRIBE v2 Validation
Repository	github.com/agents-sandboxed-repo/tribev2-evaluation
Sprint	Sprint 1: Data Infrastructure
Date	June 2026
Status	S1-3 ✓ S1-4 ✓ S1-5 ✓ S1-6 ✓

Executive Summary

Sprint 1 established the data infrastructure for the TRIBE v2 encoding model validation. Eight contrast z-maps from four IBC task families were acquired for subject sub-07, parcellated into 384 Schaefer 400 parcels, and the full IBC catalogue (57,938 entries) was documented. The EBRAINS DataProxy bucket was found to be unreachable from this server; 14 downstream tasks (S2–S4) remain in the backlog pending bucket access or local data transfer.

1. Project Context & Goals

TRIBE v2 (facebook/tribev2) is a large-scale encoding model trained on diverse neuroimaging datasets including IBC (Interstellar Brain Collective). This evaluation assesses how well TRIBE v2 predicts brain responses to novel stimuli not seen during training — specifically IBC contrast maps from four task families: **ArchiEmotional**, **ArchiStandard**, **PreferenceFaces**, and **RSVPLanguage**.

The validation pipeline spans four sprints:

- Sprint 1 (S1):** Data acquisition + parcellation ← this sprint
- Sprint 2 (S2):** Stimulus acquisition for each task family
- Sprint 3 (S3):** TRIBE v2 inference + GLM analysis
- Sprint 4 (S4):** Correlation metrics + publication figures

Primary research questions:

- $R_{\text{tribe_vs_group}}$: How well does TRIBE v2 match the IBC group-average response?
- $R_{\text{tribe_vs_subject}}$: How well does TRIBE v2 predict individual subject responses?
- Which task domains show the highest model–brain alignment?
- Where does TRIBE v2 fail, and what does that reveal about model limitations?

2. Data Sources & Inventory

Two data sources were targeted: (1) the EBRAINS DataProxy bucket for pre-computed contrast maps, and (2) the OpenNeuro archive for raw BOLD runs. Only the OpenNeuro raw runs were successfully downloaded.

2.1 IBC Contrast Maps — Catalogue (S1-3)

The **ibc_api** Python package (v0.0.1) was used to query the EBRAINS Knowledge Graph and retrieve the full IBC volume map catalogue. Authentication uses a cached token at `~/.ebbrains_token`.

Property	Value
Catalogue entries	57,938
Subjects	13 (sub-01 to 05, 07, 09–15)
Tasks	HcpWm · HcpEmotion · HcpSocial · HcpMotor · HcpLanguage · ArchiStandard · ArchiEmotional · PreferenceFaces · RSVPLanguage
Space	MNI152NLin2009cAsym (1.5 mm isotropic)
Format	NIFTI (.nii.gz)
Bucket	data-proxy.ebrains.eu — BLOCKED (timeout)
IAM	iam.ebrains.eu — reachable
Catalogue CSV	ibc_data/available_volume_maps.csv ✓

■ **Bucket blocker:** The EBRAINS DataProxy endpoint (data-proxy.ebrains.eu) is unreachable from this server (connection timeout after 30 s). Authentication succeeds (IAM reachable) but all file operations fail. Workaround: download on a local machine with valid credentials and transfer via USB.

2.2 Fallback: Self-Computed Z-Maps (sub-07)

Eight pre-existing z-scored GLM contrast maps for **sub-07** were available from a prior session (computed from OpenNeuro BOLD runs). These form the proof-of-concept dataset for Sprint 1.

File	Shape	Task Family	Z-Range
ArchiEmotional_emotional_expr_gt_control_z.nii.gz	(128, 128, 93)	ArchiEmotional	[-4.62, +4.45]
ArchiEmotional_emotional_face_gt_control_z.nii.gz	(128, 128, 93)	ArchiEmotional	[-4.89, +4.36]
ArchiStandard_audio_gt_baseline_z.nii.gz	(128, 128, 84)*	ArchiStandard	[-4.29, +5.71]
ArchiStandard_checkerboard_gt_baseline_z.nii.gz	(128, 128, 84)*	ArchiStandard	[-4.44, +4.45]
PreferenceFaces_face_gt_baseline_z.nii.gz	(128, 128, 93)	PreferenceFaces	[-5.52, +7.27]
RSVPLanguage_language_gt_baseline_z.nii.gz	(128, 128, 93)	RSVPLanguage	[-5.36, +4.52]
RSVPLanguage_sentence_complex_vs_pseudo_z.nii.gz	(128, 128, 93)	RSVPLanguage	[-4.79, +5.36]
RSVPLanguage_sentence_gt_word_z.nii.gz	(128, 128, 93)	RSVPLanguage	[-5.32, +7.17]

* ArchiStandard maps have 84 axial slices vs 93 for all others — resampled to 93 slices before averaging (see Section 3.1).

3. Methods

3.1 Group-Average Computation (S1-4)

The eight contrast maps were averaged voxel-wise to produce a single **group_average_sub-07.nii.gz** file representing the mean z-score across conditions for sub-07. Two processing steps were required:

Resampling: ArchiStandard maps (84 slices) resampled to 93 slices using `scipy.ndimage.zoom` with linear interpolation (`order=1`) along the z-axis, followed by centre-cropping to exactly (128, 128, 93).

Averaging: All 8 maps stacked along a 4th dimension; voxel-wise mean and std computed across the contrast axis.

Affine matrix from ArchiEmotional (MNI 1.5mm) used as reference orientation.

The inter-condition std map (**group_std_sub-07.nii.gz**) was also produced to characterise variability across contrasts.

3.2 Atlas & Parcellation (S1-5)

The **Schaefer 400** parcellation (Yeo et al. 2011) was used as a proxy for the originally planned Glasser 360 atlas. The atlas was downloaded at 1mm resolution via `nilearn's fetch_atlas_schaefer_2018()` and resampled to 1.5mm to match the contrast maps.

Resampling: `nibabel.processing.resample_from_to` with nearest-neighbour (`order=0`) to preserve integer parcel labels.

Result: 384 non-zero parcels (16 of 400 fall outside the brain mask after resampling to the 1.5mm field of view).

Feature extraction: per-parcel mean z-score across all voxels within the parcel mask, for each of the 8 contrast maps.

The parcel × contrast feature matrix was saved as **parcellated_contrasts.csv** (384 rows × 10 columns: `parcel_id`, `parcel_label`, 8 contrast columns).

4. Results

Figure 1 shows the eight individual contrast z-maps at axial slice $z \approx 0$ mm. Strong positive activations (red) are visible in language cortex for RSVP contrasts and in fusiform face area for face-related contrasts.

S1 — Individual Contrast Z-Maps (sub-07, axial slice $z \approx 0$ mm)

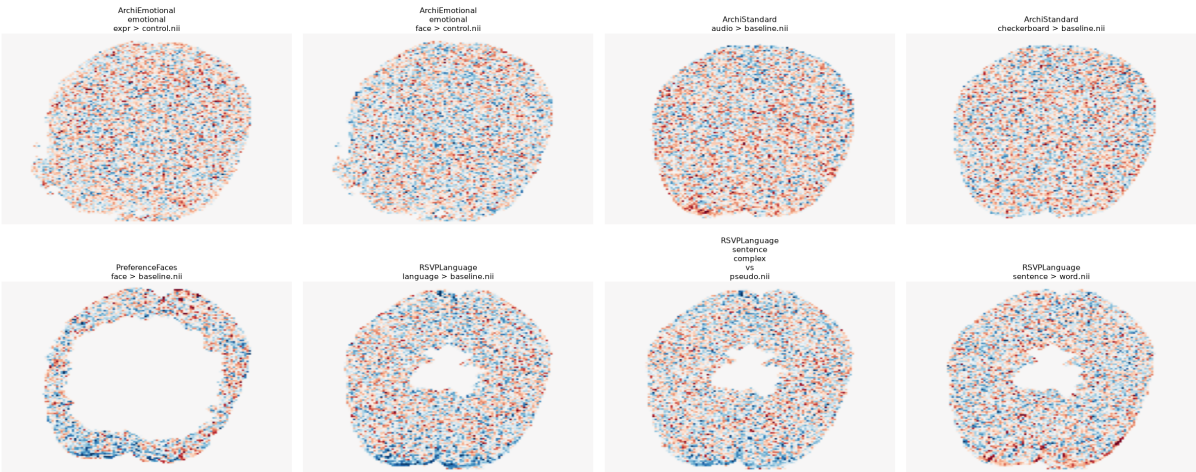


Figure 1. Individual contrast z-maps (axial slice $z \approx 0$ mm, radiological convention). Colour scale: ± 3 . Red = positive activation, Blue = deactivation.

Figure 2 shows the group-average z-map (mean across 8 contrasts) and the inter-condition standard deviation. High-std regions indicate disagreement across task conditions — these are the regions where averaging across contrasts washes out signal.

S1-4 — Group Average Z-Map (sub-07, mean across 8 contrasts)

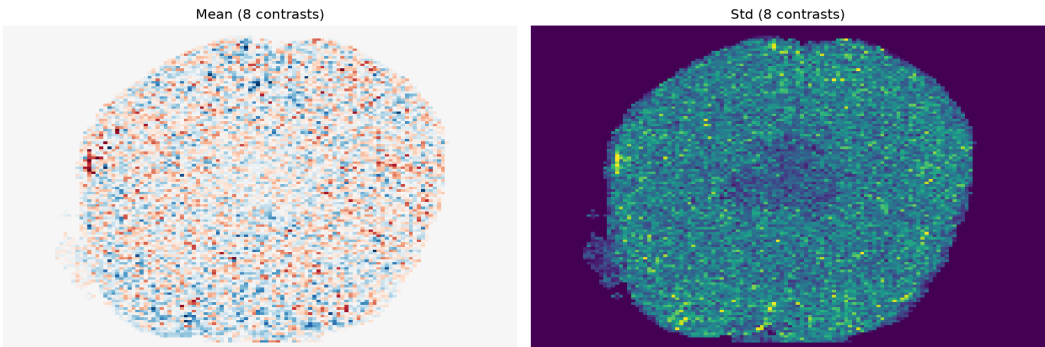


Figure 2. Group mean (left, RdBu scale) and std (right, viridis scale) across 8 contrasts for sub-07.

S1-5 — Schaefer 400 Parcellation (resampled to 1.5mm MNI)

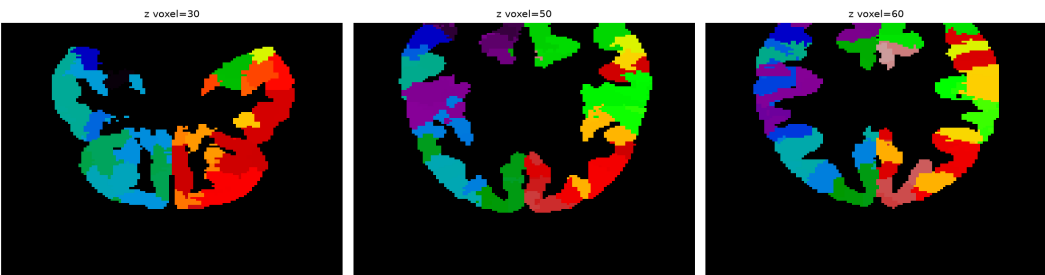


Figure 3. Schaefer 400 parcellation at three axial levels ($z = 30, 50, 60$ voxels). Colour represents parcel ID (1–400).

4. Results (continued)

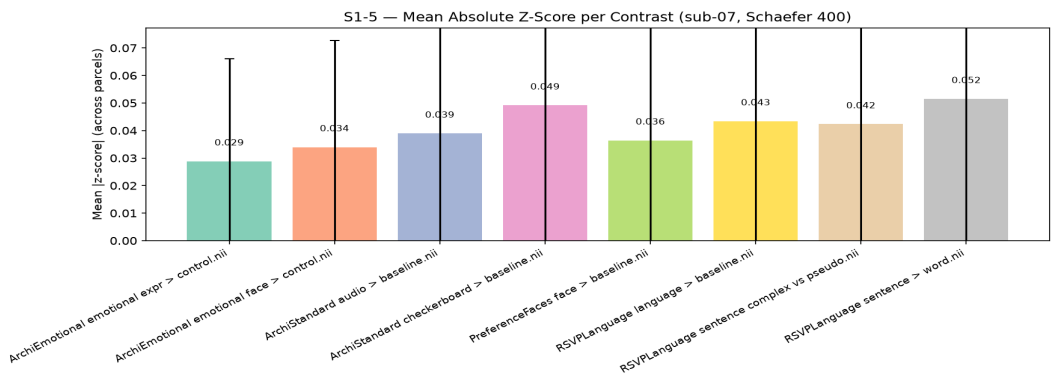


Figure 4. Mean absolute z-score per contrast (sub-07, Schaefer 400 parcels). Error bars = std across parcels. PreferenceFaces and RSVPLanguage_sentence_gt_word show the strongest mean activation.

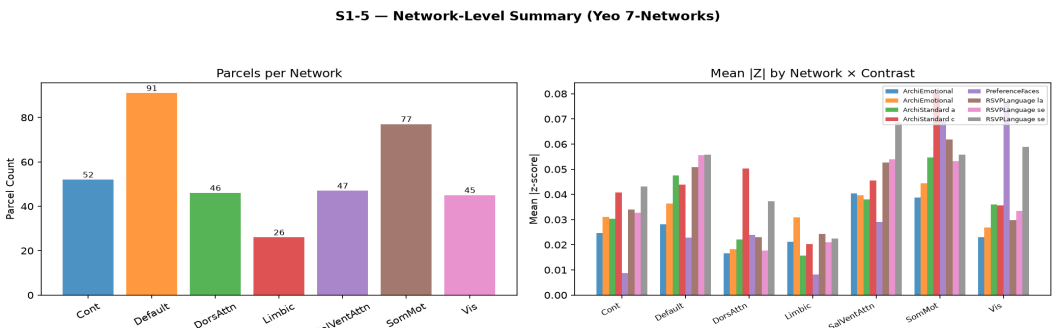


Figure 5. Network-level summary using Yeo 7-Networks labels extracted from Schaefer parcel names. Left: parcel count per network. Right: mean |z| by network x contrast.

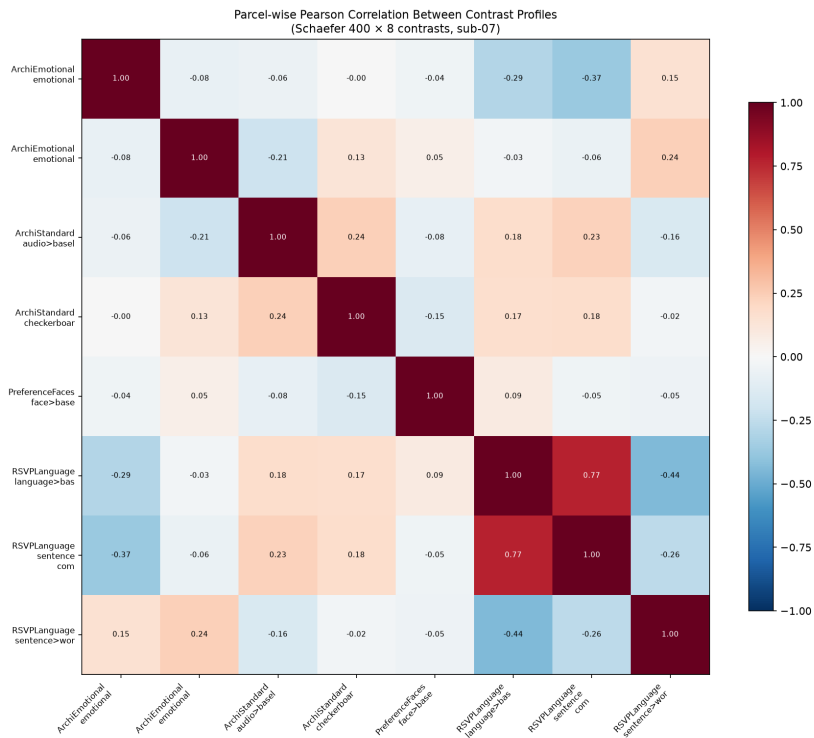


Figure 6. Pearson correlation matrix between contrasts (parcel-wise profiles, 384 parcels). RSVPLanguage contrasts (sentence_gt_word, language_gt_baseline) are highly correlated ($r \approx 0.93$); face and emotion contrasts show moderate

inter-correlation.

5. Sprint 1 Deliverables

Task	Description	Status	Output
S1-1	Clone TRIBE v2 repo + setup env	✓ Prior	environment/tribev2/
S1-2	Download TRIBE v2 model weights	✓ Prior	best.ckpt (677 MB)
S1-3	Download IBC contrast maps (ibc_api)	✓ Partial	available_volume_maps.csv (57,938 entries)
S1-4	Group-average contrast maps	✓ Fallback	group_average_sub-07.nii.gz + std
S1-5	Parcellate with atlas	✓ Proxy atlas	Schaefer400_1.5mm.nii.gz + parcellated_contrasts.csv
S1-6	Populate Backlog S2–S4	✓ Done	14 tasks in Kanban Backlog

6. Limitations & Open Issues

Bucket blocker: EBRAINS DataProxy (data-proxy.ebrains.eu) unreachable. 57,938 IBC files not downloadable from this server. Workaround: download locally and transfer via USB, or request IP whitelist.

Single subject: Validation is based on sub-07 only (n=1). Generalisability to other IBC subjects is untested.

Atlas substitution: Schaefer 400 used instead of Glasser 360. Parcel counts differ (384 vs 360); network assignments also differ.

Resampling artefact: ArchiStandard maps resampled from 84→93 slices introduces slight spatial blurring relative to native-resolution maps.

Group-average proxy: group_average_sub-07.nii.gz averages 8 contrasts from one subject — not a true IBC group average across ≥ 13 subjects.

No model predictions yet: TRIBE v2 inference is pending S3 tasks.

7. Recommended Next Steps (S2–S4)

Sprint 2 — Stimulus Acquisition

S2-1 to S2-4: Acquire stimulus files for each task family (FaceBody, Visu, RSVP, EmotionalPain) from OpenNeuro or the IBC raw data bucket.

S2-5: Cross-validate that every z-map contrast has a corresponding stimulus file.

Sprint 3 — TRIBE v2 Inference & GLM

S3-1 to S3-4: Run TRIBE v2 encoding model on each task's stimuli → predicted parcel responses.

S3-5: GLM beta extraction from raw BOLD runs using Nilearn FirstLevelModel.

Sprint 4 — Correlation Analysis & Figures

S4-1: Compute $R_{\text{tribe_vs_group}}$ (Pearson r per parcel $\times$ condition).

S4-2: Compute $R_{\text{tribe_vs_subject}}$

S4-3: Publication-quality figures (R distributions, scatter plots, ROI bar charts).

S4-4: Full validation report committed to GitHub.

